

SI Units and Unit Conversions

Answer key with worked solutions

A. Units and prefixes

1. SI base units.

(a) kilogram, **kg** (b) metre, **m** (c) second, **s** (d) kelvin, **K** (e) mole, **mol**

2. Powers of ten.

(a) $1\ \mu\text{g} = 10^{-6}\ \text{g}$ (b) $1\ \text{GHz} = 10^9\ \text{Hz}$ (c) $1\ \text{nm} = 10^{-9}\ \text{m}$

B. One-step conversions

3. Convert.

(a) $2.5\ \cancel{\text{km}} \times \frac{1000\ \text{m}}{1\ \cancel{\text{km}}} = 2.5 \times 10^3\ \text{m}$ (2500 m)

(b) $350\ \cancel{\text{mL}} \times \frac{1\ \text{L}}{1000\ \cancel{\text{mL}}} = 0.350\ \text{L}$

(c) $45\ \cancel{\text{mg}} \times \frac{1\ \text{g}}{1000\ \cancel{\text{mg}}} = 0.045\ \text{g}$

(d) $0.0045\ \cancel{\text{kg}} \times \frac{1000\ \cancel{\text{g}}}{1\ \cancel{\text{kg}}} \times \frac{1000\ \text{mg}}{1\ \cancel{\text{g}}} = 4.5 \times 10^3\ \text{mg}$

(e) $500.\ \cancel{\text{nm}} \times \frac{10^{-9}\ \text{m}}{1\ \cancel{\text{nm}}} = 5.00 \times 10^{-7}\ \text{m}$

C. Areas and volumes

4. Convert.

(a) $5.0\ \cancel{\text{m}^2} \times \frac{10^4\ \text{cm}^2}{1\ \cancel{\text{m}^2}} = 5.0 \times 10^4\ \text{cm}^2$ (because $1\ \text{m}^2 = (100\ \text{cm})^2$)

(b) $250\ \cancel{\text{cm}^3} \times \frac{1\ \text{m}^3}{10^6\ \cancel{\text{cm}^3}} = 2.50 \times 10^{-4}\ \text{m}^3$

(c) $1.0\ \cancel{\text{m}^3} \times \frac{1000\ \text{L}}{1\ \cancel{\text{m}^3}} = 1.0 \times 10^3\ \text{L}$ ($1\ \text{m}^3 = 1000\ \text{dm}^3 = 1000\ \text{L}$)

(d) $0.250\ \cancel{\text{L}} \times \frac{1000\ \text{cm}^3}{1\ \cancel{\text{L}}} = 250.\ \text{cm}^3$

D. Multi-step conversions

5. 1.5 h.

$$1.5 \cancel{\text{h}} \times \frac{3600 \text{ s}}{1 \cancel{\text{h}}} = \mathbf{5.4 \times 10^3 \text{ s}}$$

6. 36 km/h.

$$\frac{36 \cancel{\text{km}}}{1 \cancel{\text{h}}} \times \frac{1000 \text{ m}}{1 \cancel{\text{km}}} \times \frac{1 \cancel{\text{h}}}{3600 \text{ s}} \\ = \mathbf{10 \text{ m/s}}$$

7. Iron.

$$\frac{7.86 \cancel{\text{g}}}{1 \cancel{\text{cm}^3}} \times \frac{1 \text{ kg}}{1000 \cancel{\text{g}}} \times \frac{10^6 \cancel{\text{cm}^3}}{1 \text{ m}^3} \\ = \mathbf{7.86 \times 10^3 \text{ kg/m}^3}$$

8. Mercury.

$$15.0 \cancel{\text{mL}} \times \frac{13.6 \text{ g}}{1 \cancel{\text{mL}}} = \mathbf{204 \text{ g}}$$

E. Challenge

9. Swimming pool.

$$\text{(a) } V = 25.0 \text{ m} \times 10.0 \text{ m} \times 2.00 \text{ m} = \mathbf{500. \text{ m}^3}$$

$$500. \cancel{\text{m}^3} \times \frac{1000 \text{ L}}{1 \cancel{\text{m}^3}} = \mathbf{5.00 \times 10^5 \text{ L}}$$

(b)

$$m = 5.00 \times 10^5 \cancel{\text{L}} \times \frac{1000 \cancel{\text{mL}}}{1 \cancel{\text{L}}} \\ \times \frac{1.00 \cancel{\text{g}}}{1 \cancel{\text{mL}}} \times \frac{1 \text{ kg}}{1000 \cancel{\text{g}}} \\ = \mathbf{5.00 \times 10^5 \text{ kg}}$$